

Finite-Probe Inverse-Jet Reconstruction for Pandrosion

Symmetric probe stencils, tensor recovery, and derivative-free chart flattening

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Abstract

The all-order tensor inverse-jet theorem identifies the optimal local chart for a regular zero of an analytic system, but it is stated in terms of Taylor tensors. A derivative-free Pandrosion engine should recover those tensors from finite probes. This paper formulates and proves the local consistency statement behind that goal. For an analytic map $G : \mathbb{C}^n \rightarrow \mathbb{C}^n$ near a regular zero ρ , finite symmetric probes at radius h recover the homogeneous Taylor coefficients G_q with error $\mathcal{O}(h)$, provided the probe stencil is unisolvent for homogeneous polynomials through degree m . Applying the inverse-jet recursion to the recovered tensors gives estimated chart coefficients $\hat{B}_q(h)$ satisfying

$$\hat{B}_q(h) \longrightarrow B_q, \quad 1 \leq q \leq m,$$

where B_q are the exact all-order inverse-jet coefficients. Consequently the reconstructed chart $\hat{\phi}_m$ is asymptotically m -flat:

$$G(\hat{\phi}_m(u)) = u + \mathcal{O}_h(\|u\|^{m+1}) + o(1)$$

on bounded chart coordinates. The paper closes with a finite-slope engine conjecture: stable Pandrosion probes with rank filtering and equal-value avoidance implement this reconstruction asymptotically without explicit derivative access.

Scope and status. The main results are local consistency theorems for finite probe reconstruction. They do not claim global convergence of a solver; they supply the derivative-free bridge between tensor inverse jets and practical Pandrosion chart engines.

1 Problem

The all-order inverse-jet chart at a regular zero is

$$\phi_m(u) = \rho + \sum_{q=1}^m B_q[u, \dots, u],$$

chosen so that

$$G(\phi_m(u)) = u + \mathcal{O}(\|u\|^{m+1}).$$

The coefficients B_q are computable from the Taylor tensors of G . A Pandrosion engine, however, observes values of G and finite slopes, not exact derivatives. We therefore ask:

Can the inverse-jet coefficients be reconstructed from finite probe evaluations as the probe radius tends to zero?

2 Probe stencils

Let H_q denote the finite-dimensional vector space of homogeneous polynomial maps $P : \mathbb{C}^n \rightarrow \mathbb{C}^n$ of degree q .

Definition 2.1 (Unisolvent probe stencil). *A finite set of directions $V = \{v_1, \dots, v_N\} \subset \mathbb{C}^n$ is unisolvent through degree m if, for every $1 \leq q \leq m$, the evaluation map*

$$H_q \rightarrow (\mathbb{C}^n)^N, \quad P \mapsto (P(v_1), \dots, P(v_N))$$

is injective.

Remark 2.2. *Generic directions are unisolvent once N is at least the dimension of the relevant homogeneous polynomial space. In implementation, one may use an overdetermined stencil and solve by least squares.*

Write the Taylor expansion at a regular zero ρ as

$$G(\rho + w) = \sum_{q \geq 1} G_q[w, \dots, w], \quad G_1 = DG(\rho).$$

3 Homogeneous tensor recovery

Theorem 3.1 (Finite-probe Taylor tensor recovery). *Let G be analytic near ρ , and let V be unisolvent through degree m . For $h > 0$, define homogeneous coefficient estimates recursively by fitting the values*

$$R_q(h, v_i) = h^{-q} \left(G(\rho + hv_i) - \sum_{\ell=1}^{q-1} h^\ell \hat{G}_\ell(h)[v_i, \dots, v_i] \right)$$

over the stencil V . Then, for each $1 \leq q \leq m$,

$$\hat{G}_q(h) \rightarrow G_q \quad \text{as } h \rightarrow 0.$$

If the evaluations of G are exact, the error is $\mathcal{O}(h)$.

Proof. For $q = 1$,

$$h^{-1}G(\rho + hv) = G_1[v] + \mathcal{O}(h)$$

uniformly for $v \in V$. Since the evaluation map on H_1 is injective and finite-dimensional, the inverse on its image is bounded after choosing any fixed reconstruction operator. Hence $\hat{G}_1(h) = G_1 + \mathcal{O}(h)$.

Assume $\hat{G}_\ell(h) = G_\ell + \mathcal{O}(h)$ for all $\ell < q$. Taylor expansion gives

$$G(\rho + hv) = \sum_{\ell=1}^q h^\ell G_\ell[v, \dots, v] + \mathcal{O}(h^{q+1}).$$

Subtracting the already reconstructed lower-order terms and dividing by h^q gives

$$R_q(h, v) = G_q[v, \dots, v] + \mathcal{O}(h)$$

on the finite stencil. Unisolvency again converts value error into coefficient error. Induction proves the result. $\square$

4 Inverse-jet reconstruction

The inverse-jet recursion is algebraic. Let $A = G_1$ and let B_q be the exact coefficients of the inverse jet. The first coefficient is $B_1 = A^{-1}$. At order k , the coefficient B_k solves

$$AB_k[u, \dots, u] + R_k(G_2, \dots, G_k; B_1, \dots, B_{k-1})(u) = 0,$$

where R_k is the homogeneous degree- k residual determined by lower data.

Theorem 4.1 (Consistency of finite-probe inverse jets). *Assume G_1 is invertible and the hypotheses of Theorem 3.1 hold through degree m . Let $\hat{B}_q(h)$ be obtained by applying the inverse-jet recursion to $\hat{G}_1(h), \dots, \hat{G}_q(h)$. Then*

$$\hat{B}_q(h) \rightarrow B_q, \quad 1 \leq q \leq m.$$

With exact evaluations, $\hat{B}_q(h) - B_q = \mathcal{O}(h)$ for every fixed $q \leq m$.

Proof. The map $G_1 \mapsto G_1^{-1}$ is continuous near the invertible matrix G_1 , so $\hat{B}_1(h) \rightarrow B_1$. Suppose the result holds through order $k-1$. The residual polynomial R_k is a finite algebraic expression in $G_2, \dots, G_k$ and $B_1, \dots, B_{k-1}$; therefore its finite-probe estimate converges to the exact residual. Multiplication by $\hat{G}_1(h)^{-1}$ is also continuous. Thus $\hat{B}_k(h) \rightarrow B_k$. The $\mathcal{O}(h)$ rate follows from the same induction because all algebraic maps are locally Lipschitz on bounded finite-dimensional neighborhoods with G_1 bounded away from singularity. $\square$

Corollary 4.2 (Asymptotic m -flatness). *Let*

$$\hat{\phi}_m^{(h)}(u) = \rho + \sum_{q=1}^m \hat{B}_q(h)[u, \dots, u].$$

For every fixed bounded set of u -coordinates,

$$G(\hat{\phi}_m^{(h)}(u)) = u + \mathcal{O}(\|u\|^{m+1}) + \mathcal{O}(h) \sum_{q=1}^m \|u\|^q.$$

In particular, $\hat{\phi}_m^{(h)}$ converges coefficientwise to an m -flat inverse-jet chart.

Proof. Substitute the coefficient convergence from Theorem 4.1 into the finite Taylor expansion of G . The exact inverse jet cancels homogeneous terms through degree m , and the coefficient error contributes the stated lower-order perturbation. $\square$

5 Noise and conditioning

Proposition 5.1 (Stable overdetermined reconstruction). *Suppose the stencil reconstruction operator has condition number at most κ_V , and suppose probe evaluations have additive error at most ε_h . Then the recovered q -th Taylor tensor has an additional error bounded by*

$$\mathcal{O}(\kappa_V \varepsilon_h h^{-q})$$

at order q . Consequently consistent order- m reconstruction requires $\varepsilon_h = o(h^m)$ for the highest tensor.

Proof. The fit at degree q is a finite-dimensional linear inverse problem after lower-order subtraction. Additive value errors are divided by h^q , and the reconstruction operator amplifies them by at most κ_V . The highest recovered degree imposes the strongest scaling requirement. $\square$

6 Pandrosion engine conjecture

Conjecture 6.1 (Derivative-free inverse-jet reconstruction by stable probes). *A Pandrosion engine using symmetric probe clouds, finite-slope rank filtering, equal-value pole avoidance, and shrinking probe radius h reconstructs the all-order inverse-jet coefficients through degree m whenever the probe stencil remains uniformly unisolvent and the effective evaluation noise is $o(h^m)$.*

Conjecture 6.2 (Reconstructed chart multiplier law). *Under the hypotheses above, the charted anchored Pandrosion map built from the reconstructed chart satisfies*

$$\left\| DP_{G \circ \widehat{\phi}_m^{(h)}, a}(0) \right\| \leq C \|a\|^m + o(1)$$

as $h \rightarrow 0$, uniformly for anchors in a fixed small chart ball.

7 Conclusion

The all-order inverse jet is not merely a formal derivative object. It can be recovered from finite probes, order by order, provided the stencil is rich enough and the probe radius is controlled. This makes the m -flat atlas programme algorithmic: exact derivatives define the target, while symmetric finite probes supply a derivative-free path toward it.

References

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